

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 24: Solution of Right Triangles

Every formula and result of the chapter on one sheet

1	Perpendicular	$\text{Hypotenuse} \times \sin \theta$
2	Base	$\text{Hypotenuse} \times \cos \theta$
3	Perpendicular from the base	$\text{Base} \times \tan \theta$
4	Hypotenuse	$\frac{\text{perpendicular}}{\sin \theta} = \frac{\text{base}}{\cos \theta}$
5	Acute angles	$\angle A + \angle C = 90^\circ$
6	30-60-90 triangle	$1 : \sqrt{3} : 2$
7	45-45-90 triangle	$1 : 1 : \sqrt{2}$
8	Side opposite 30°	Half the hypotenuse
9	Diagonal of a square	$\text{Side} \times \sqrt{2}$
10	Altitude of an equilateral triangle	$\frac{\sqrt{3}}{2} \times \text{side}$
11	Chord subtending 2θ at the centre	$2r \sin \theta$
12	Surd values	$\sqrt{2} \approx 1.414, \sqrt{3} \approx 1.732$